

Revisiting completeness of the eigenstates of a Hermitian operator for a particle in 1-D: an alternative proof, an alternative description, and the possibility of quantum measurement of the momentum in a box

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(Dated: July 14, 2026)

We have revisited the completeness in quantum measurement and derived the closure relation ($\frac{2}{a} \sum_{n=1}^{\infty} \sin(n\pi x/a) \sin(n\pi x'/a) = \delta(x - x')$) for completeness of the energy eigenfunctions by the use of the Poisson summation formula but, without the use of basis expansion property $f(x) = \sum_n c_n \psi_n(x)$ for a quantum particle in a 1-D box of length a . This derivation is, essentially, an alternative proof of completeness. The momentum operator, however, is neither self-adjoint nor bound in the usual Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ for the particle in the box geometry. It raises a question whether momentum of the particle in the box is really measurable. The straightforward answer, of course, is ‘yes’. We have found a critical answer to this question from the consideration of completeness of the momentum eigenfunctions ($\in \mathcal{H}_{\infty} = \mathcal{L}^2((-\infty, \infty), w(x) = 1, dx)$) of the particle in the 1-D box. We have also proposed an alternative way of describing completeness of the eigenfunctions of a Hermitian operator in a very simple algebraic manner, say, $\psi_m(x) = \sum_n \psi_n(x) \delta_{n,m}$ for all $\psi_m(x) \in \mathcal{L}^2((a, b), w(x) = 1, dx)$. The alternative derivation, the critical answer, and the proposition offer a deeper understanding of completeness in quantum measurement.

I. INTRODUCTION

The idea of complete set of the eigenstates of a Hermitian operator is one of the fundamental postulates of quantum mechanics [1]. The idea was put forth by Dirac in connection with the formulation of quantum mechanics in 1930 [1]. In connection with the same, Dirac stated – “any quantity that can be measured is an observable” and “we call a real dynamical variable whose eigenstates form a complete set an *observable*” [1]. So, the quantum (theoretical) measurement of any physical quantity requires a complete set of eigenstates of the observable. This makes the idea of completeness necessary and ubiquitous in quantum mechanics [2–5].

The idea of completeness is one of the fundamental postulates (or axioms) of quantum mechanics [6]. Consequently, the energy eigenstates of a particle are complete, as because its energy (or the corresponding Hamiltonian) is an observable [1]. Consequently, the energy eigenfunctions (eigenstates) form an orthonormal basis according to the Sturm-Liouville theory [7]. Since the quantum measurement, even for a quantum dot (which mimics a particle in a box) [8], is still relevant, completeness of the energy eigenfunctions for a particle in a 1-D box is also relevant in the current context. Box-potential has also been experimentally realized somewhat recently as a magneto-optical trap [9]. The study of physical properties of the Bose-Einstein condensate [10–13] in such a box-trap trap is still a growing topic from both the theo-

retical point of view [14–16] and the experimental point of view [9].

Completeness of a set of orthonormal functions $\{\psi_n(x)\}$ in the Hilbert space, say $\mathcal{H} = \mathcal{L}^2((a, b), w(x) = 1, dx)$ ¹ for a 1-D problem [17], however, is often introduced as a property that an arbitrary function $f(x) \in \mathcal{H}$ can be expressed as a linear combination of all the orthonormal and complete set of eigenfunctions such that²

$$f(x) = \sum_n c_n \psi_n(x) \quad (1)$$

with the constant coefficient³ $c_n = \int_a^b \psi_n^*(x) f(x) dx$ for all x in the interval (a, b) [18, 19]. In connection with the same, especially for the energy eigenvalue problem of the particle in a 1-D box (infinite square well) of length a and complete set of the energy eigenfunctions

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad (2)$$

in the interval $[0, a]$ with the Dirichlet boundary conditions for $n = 1, 2, 3, \dots$, Griffiths in his textbook on *introduction to quantum mechanics*, mentioned“

$$f(x) = \sum_{n=1}^{\infty} c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad (3)$$

¹ Here, $\mathcal{L}^2((a, b), w(x) = 1, dx)$ stands for Lebesgue space for inner products of square-integrable functions (of x defined in the interval (a, b)) with the weight function $w(x) = 1$.

² Wherever the lower and upper limits of the summation over n are not specified, it indicates summation over the complete set.

³ Here, c_n is the probability amplitude of $\psi_n(x)$ in $f(x)$.

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is nothing but the Fourier series of $f(x)$ " [2, 3] for an arbitrary $f(x) \in \mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ ⁴. He further added that the convergence of the above sine series to $f(x)$ with

$$c_n = \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x'}{a}\right) f(x') dx' \quad (4)$$

can be proved within Dirichlet's theorem for the Fourier series [2, 3]. This convergence is coined as a proof of completeness of the energy eigenfunctions for a particle in a 1-D box [6]. Such a proof of completeness, of course, is not directly possible for a general case of the time-independent Schrödinger equation [6].

The applicability of Dirichlet's theorem for the Fourier series, of course, needs the function $f(x)$ (as mentioned in Eqn. (3)) to be periodic in $\mathbb{R}$. Since $f(x) \in \mathcal{H}$ is defined only in the interval $(0, a)$ without any mention of the periodicity (or periodic extension), it is not clear how one can apply Dirichlet's theorem to Eqn. (3). A clarification, in this regard, may be described as follows. The right-hand side (RHS) of Eqn. (3), however, could be a Fourier series of a $2a$ -periodic and odd function $F(x)$ with the properties $F(x) = f(x)$ only for all $x \in (0, a)$, and $F(-x) = -F(x)$ and $F(x) = F(x + 2a)$ for all $x \in \mathbb{R}$. Thus, according to Dirichlet's theorem for the Fourier series of $F(x)$, the RHS of Eqn. (3) converges to $f(x)$ for all $x \in (0, a)$ with the same c_n as mentioned in Eqn. (4). However, we would like to offer a simple alternative proof of the same by deriving the closure relation for completeness of the energy eigenfunction without the direct use of the Dirichlet theorem.

In the literature, the closure relation for completeness of the energy eigenfunctions (which form a complete and orthonormal basis) is easily obtained from its matrix form (closure) $\sum_n |\psi_n\rangle \langle \psi_n| = \mathbb{1}$ [20], which follows from the property, $|f\rangle = \sum_n c_n |\psi_n\rangle$ (with $c_n = \langle \psi_n | f \rangle$), of the orthonormal and complete set of the energy eigenkets in a subset of the Hilbert space where the arbitrary state $|f\rangle$ belongs to. Here, the basis expansion property $|f\rangle = \sum_n c_n |\psi_n\rangle$ is treated as an axiom for

the completeness of $\{|\psi_n\rangle\}$ [5]. The matrix element $\langle x | (\sum_n |\psi_n\rangle \langle \psi_n|) | x' \rangle = \langle x | \mathbb{1} | x' \rangle$ in both sides of the closure with Dirac normalization $\langle x | x' \rangle = \delta(x - x')$ results in the desired closure relation

$$\sum_n \psi_n(x) \psi_n^*(x') = \delta(x - x') \quad (5)$$

for the energy eigenfunctions [4]. We would, however, like to illustrate an alternative derivation of the closure relation (Eqn. (5)) for completeness of the energy eigenfunctions for the particle in the 1-D box problem without considering the basis expansion property (Eqn. (1)) as an axiom [6].

The momentum operator, however, is neither self-adjoint nor bound in the usual Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ for the particle in the box geometry [22–24]. Consequently, the momentum operator $\hat{p}_x = -i\hbar \frac{d}{dx}$ is not necessarily a Hermitian operator on the Hilbert space $\mathcal{H}$. It would have been a Hermitian one only in a sub-domain of $\mathcal{H}$, where all the functions obey periodic (including the Dirichlet) boundary conditions [22]. Since the Hilbert space $\mathcal{H}$ is a complete inner-product space, it contains all kinds of square-integrable functions, including the ones which obey the Neumann boundary conditions, such as $f(x) = \sqrt{2/a} \cos(\pi x/a)$. Thus, it raises a question whether momentum of the particle in the box is really measurable. The straightforward answer is 'yes'. We, however, would like to critically answer this question from the completeness of the momentum eigenstates point of view. It shortcoming with the momentum operator though, have been discussed in various literature [22–24], a rigorous explanation of the possibility of the quantum measurement of the momentum operator $\hat{p}_x = -i\hbar \frac{d}{dx}$, as far as we know, is not found there.

The eigenstates $\{\psi_n(x)\} \in \mathcal{H} = \mathcal{L}^2((a, b), w(x) = 1, dx)$ of a Hermitian operator form an orthonormal set of basis functions in the Hilbert space $\{\psi_n(x)\} \in \mathcal{H}$ according to the Sturm-Liouville theory, even for a general 1-D problem [7]. Completeness of an orthonormal set of the basis functions ($\{\psi_n(x)\} \in \mathcal{H}$) is either introduced either as a basis expansion property (Eqn. (1) [18, 19] or as the closure relation (Eqn. (5) [4]), as mentioned above. It is easy to go from one to another and vice versa. We, however, would like to propose another alternative form $\psi_m(x) = \sum_n \psi_n(x) \delta_{n,m}$ (for all $\psi_m(x) \in \mathcal{L}^2((a, b), w(x) = 1, dx)$) of the same in a very simple algebraic manner. Such a simple form, though powerful, is also not found in the literature.

In this *article*, our derivation starts in Section 2 A with the Fourier series expansion of the a -periodic Dirac-comb and the Poisson summation formula (PSF) [25, 26]. We use this formula to obtain the closure relation for completeness of the energy eigenfunctions for the particle in a 1-D box. In Section 2 A, we critically and rigorously answer the question whether momentum of the particle

⁴ Note that for the particle in the 1-D box problem, the boundary points ($x = 0$ and $x = a$) are excluded in the expression of the Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$, as because a subset of it is the domain of the Hamiltonian operator $\hat{H}_x = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$ which is well defined with the potential energy $V(x) = 0$ only in $x \in (0, a)$ [21]. The Hamiltonian operator, of course, is not well defined beyond the interval $\in (0, a)$ as potential energy $V(x) \rightarrow \infty$ which, however, results in the Dirichlet boundary conditions ($\psi_n(0) = \psi_n(a) = 0$) on the energy eigenfunctions. The kinetic energy operator $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ is, of course, undefined at the boundary points. The Hamiltonian operator, being Hermitian on a subset of $\mathcal{H}$ (say, $\mathcal{H}' \subset \mathcal{H}$), obeys $\langle g | \hat{H}_x f \rangle = \langle \hat{H}_x g | f \rangle$ for any $f(x)$ and $g(x) \in \mathcal{H}'$ which satisfy the Dirichlet/Neumann/periodic boundary conditions (or their mixture(s)) according to the Sturm-Liouville theory [7].

in the box is quantum mechanically measurable by virtue of completeness of the momentum eigenstates. In Section 3, we propose an alternative form of completeness of the basis functions. Finally, we conclude in Section 4.

II. THE ALTERNATIVE DERIVATION OF THE CLOSURE RELATION AND THE POSSIBILITY OF MEASUREMENT OF MOMENTUM OF A PARTICLE IN A 1-D BOX

A. The alternative derivation of the closure relation for a particle in a 1-D box

Let us take an a -periodic Dirac-comb

$$g(x) = \sum_{n=-\infty}^{\infty} \delta(x/a - n) \quad (6)$$

for all $x \in (-\infty, \infty)$. The Fourier series expansion of the Dirac-comb $g(x)$ results in

$$\sum_{n=-\infty}^{\infty} \delta(x/a - n) = \sum_{n=-\infty}^{\infty} d_n e^{2\pi i n x/a}, \quad (7)$$

where all the Fourier coefficients take the same value $d_n = 1$. Eqn. (7) is the PSF [25, 26] which is often used to evaluate the difference between the summation and the definite integration of a function on an interval [27–29]. It should be mentioned in this regard that the Fourier transform of a Dirac-comb in the position space is also a Dirac-comb in the momentum space. One can directly show this by applying the PSF [30].

It is important to note that Dirichlet's theorem for the Fourier series fails to prove the convergence of the Fourier series on the RHS of Eqn. (7) to the left-hand side (LHS) of the same, as because the Riemann–Lebesgue lemma⁵ is not satisfied for this particular case of the Dirac-comb $g(x)$, which, of course, doesn't have the required bounded variation [31] in the interval $[-a/2, a/2]$ [32, 33]. Hence, Dirichlet's theorem fails to prove completeness of the Fourier modes $\{e^{2\pi i n x/a}\}$ (with $n = 0, \pm 1, \pm 2, \dots$) in the Dirac-comb $g(x)$. The convergence of the RHS of Eqn. (7) to the LHS of the same can, however, be proved by virtue of the properties of Dirichlet's kernel $D_N(x) = \sum_{n=-N}^N e^{2\pi i n x/a} = \frac{\sin([1+2N]\pi x/a)}{\sin(\pi x/a)}$, which is nothing but a δ -sequence function [34] near around every $x \in \mathbb{N}$ and it becomes $\delta(x/a - n)$ for all $x \in [an - a/2, an + a/2]$ in the limiting case of $N \rightarrow \infty$ [32, 33]. We, however,

will directly derive the closure relation for completeness of the energy eigenfunctions from the PSF.

For the particle in a 1-D box problem, according to Eqn. (2), we have

$$\sum_n \psi_n(x) \psi_n^*(x') = \frac{2}{a} \sum_{n=1}^{\infty} \sin(n\pi x/a) \sin(n\pi x'/a) \quad (8)$$

for all x and x' in the interval $(0, a)$. Let us replace the sinusoidal functions of Eqn. (8), say $\sin(n\pi x/a)$, with the exponential forms, say $[e^{in\pi x/a} - e^{-in\pi x/a}]/2i$, and further replace the exponential forms with the Dirac deltas according to Eqn. (7), to get

$$\begin{aligned} \frac{2}{a} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi x'}{a}\right) &= \frac{1}{2a} \left[\sum_{n=-\infty}^{\infty} \delta\left(\frac{x-x'}{2a} - n\right) \right. \\ &\quad \left. - \sum_{n=-\infty}^{\infty} \delta\left(\frac{x+x'}{2a} - n\right) \right]. \end{aligned} \quad (9)$$

Eqn. (9) describes the key result towards the alternative derivation of the closure relation. It is interesting to note that for the 1st summation on the RHS of Eqn. (9), all the Dirac deltas are 0 except the one with $n = 0$ because of $-\frac{1}{2} < \frac{x-x'}{2a} < \frac{1}{2}$. Similarly, for the 2nd summation on the RHS of the same equation, all the Dirac deltas are 0 because of $0 < \frac{x+x'}{2a} < 1$. So, essentially, we have only one effective Dirac delta out of infinite ones in Eqn. (9). Thus, we have $\frac{2}{a} \sum_{n=1}^{\infty} \sin(n\pi x/a) \sin(n\pi x'/a) = \frac{1}{2a} \delta\left(\frac{x-x'}{2a} - 0\right)$ which can be further recast according to a property of the Dirac delta, say $\delta(bx) = \frac{1}{|b|} \delta(x)$, as

$$\frac{2}{a} \sum_{n=1}^{\infty} \sin(n\pi x/a) \sin(n\pi x'/a) = \delta(x - x'). \quad (10)$$

This is the closure relation for completeness of the energy functions of a particle in a 1-D box obtained from the PSF. Eqn. (10) completes the desired alternative derivation. The derived closure relation (Eqn. (10)) essentially is proof completeness of the energy eigenfunctions, as because the proof is now straightforward.

The RHS of Eqn. (3) can now be recast with Eqns. (4) and (10), as

$$\begin{aligned} &\sum_{n=1}^{\infty} c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \\ &= \int_0^a f(x') \left[\frac{2}{a} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi x'}{a}\right) \right] dx' \\ &= \int_0^a f(x') \delta(x - x') dx' = f(x) \end{aligned} \quad (11)$$

for all $x \in (0, a)$. Eqn. (11) concludes proof of the convergence of the RHS of Eqn. (3) to $f(x) \in \mathcal{H} =$

⁵ According to the Riemann–Lebesgue lemma, one expects $\lim_{n \rightarrow \pm\infty} d_n = 0$ for an a -periodic function having bounded variation in the interval $[-a/2, a/2]$.

$\mathcal{L}^2((0, a), w(x) = 1, dx)$. Proof of the convergence of the sine series to $f(x)$, as described above, is applicable for any (square-integrable⁶) continuous function $f(x) \in \mathcal{H}$ in the interval $(0, a)$. Here, the convergence is shown to be of point-wise type for every $x \in (0, a)$ by virtue of the continuity of $f(x')$ and the use of Eqn. (10) involving the Dirac delta $\delta(x - x')$ for all $x' \in (0, a)$. Eqn. (11) is proof of completeness of the energy eigenfunctions for the particle in the 1-D box.

B. The possibility of the quantum measurement of momentum of the particle in the box

Here, for the particle in the 1-D box problem, a subset of the Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ represents domain of the Hamiltonian operator ($\hat{H}_x = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$) which is well-defined only in the interval $(0, a)$, and it acts on the energy eigenfunctions $\{\psi_n(x), n = 1, 2, 3, \dots\}$ ⁷ that are defined in the interval $[0, a]$ with the Dirichlet boundary conditions $\psi_n(0) = \psi_n(a) = 0$ [21]. The Hilbert space, being a complete inner-product space, accommodates all possible square-integrable functions $\{f(x)\}$ which are defined at least in the interval $(0, a)$. It is, however, not expected that all such functions satisfy the same (Dirichlet) boundary conditions as satisfied by the energy eigenfunctions $\{\psi_n(x) = \sqrt{\frac{2}{a}} \sin(\frac{n\pi x}{a})\}$. Even a cosine function, say $f(x) = \sqrt{\frac{2}{a}} \cos(\frac{\pi x}{a}) = \sum_{n=1}^{\infty} c_j \sqrt{\frac{2}{a}} \sin(\frac{j\pi x}{a})$ with different (Neumann) boundary conditions as illustrated in inset-a of figure 1 for the summation over j from $j = 1$ to $j = 50$, can be accommodated in $\mathcal{H}$ for all $x \in (0, a)$ with the coefficients $c_j = \frac{2j[1+\cos(j\pi)]}{[j^2-1]\pi}$ ⁸ (where $j = 1, 2, 3, \dots$ and $b_1 = 0$)⁹ by virtue of the completeness of the energy eigenfunctions. A Hermitian operator can also operate on such a function. Thus, the concept of completeness for the quantum measurement is still applicable for the basis (energy eigenfunctions) expansion of such functions, as we haven't considered any specific boundary conditions for $f(x)$ in Section 2 A.

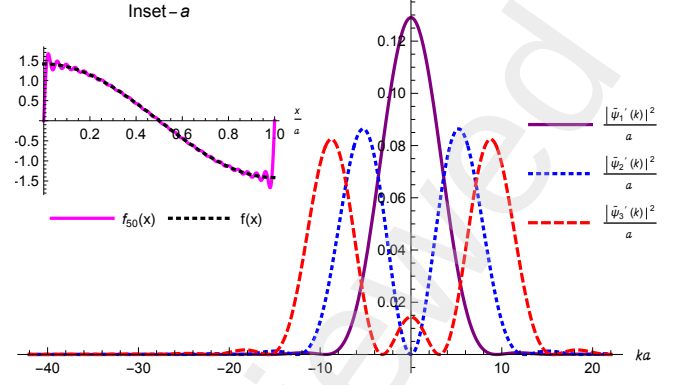


FIG. 1: The plots represent the normalized probability distribution ($\{|\psi'_n(k)|^2\}$) of the quantum measurement of momentum with the eigenvalue $p_x = \hbar k$ in the states $\{\psi'_n(x); n = 1$ (solid line), 2 (dotted line), & 3 (dashed line)} and follow Eqn. (13). In inset-a, the dotted line represents $f(x) = \sqrt{\frac{2}{a}} \cos(\pi x/a)$ and the solid line represents the basis expansion with the first 50 modes ($f_{50}(x) = \sum_{j=1}^{50} c_j \sqrt{\frac{2}{a}} \sin(j\pi x/a)$) of the same (as introduced in Eqn. (3)) for $a = \text{unity}$.

The result $(-i\frac{n\pi}{a}\sqrt{2/a}\cos(n\pi x/L))$ of the operation $\hat{p}_x \psi_n(x)$ though belongs to the Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ as indicated in inset-a of figure 1, the quantum measurement of momentum is not possible there ($\mathcal{H}$). The Hilbert space $\mathcal{H}$, of course, doesn't have the necessary domain for the momentum operator $\hat{p}_x = -i\hbar \frac{d}{dx}$ as it is neither bound nor self-adjoint in $\mathcal{H}$ [22–24]. Consequently, the so-called momentum eigenstates $\frac{1}{\sqrt{2\pi}}e^{ikx}$ and $\frac{1}{\sqrt{2\pi}}e^{ik'x}$, for arbitrary wave-numbers $k, k' \in \mathbb{R}$, are neither orthogonal nor complete in $\mathcal{H}$. Thus, the quantum (theoretical) measurement of momentum of the particle in the 1-D box is not possible in $\mathcal{H}$.

Real measurement, however, doesn't care about the mathematical choice of the Hilbert space. The appropriate (rigged [35]) Hilbert space for the quantum (theoretical) measurement in this regard, would be $\mathcal{H}_\infty = \mathcal{L}^2((-\infty, \infty), w(x) = 1, dx)$. If momentum of the particle in the 1-D box is now measured in a relevant quantum mechanical state¹⁰

$$\psi'_n(x) = \begin{cases} \psi_n(x) & \text{for } 0 < x < a \\ 0 & \text{for all other } x \in \mathbb{R} \end{cases} \in \mathcal{H}_\infty, \quad (12)$$

then the probability amplitude corresponding to the outcome $p = \hbar k$ can be obtained from the Fourier transform

¹⁰ Here, $\psi_n(x) = \sqrt{\frac{2}{a}} \sin(\frac{n\pi x}{a})$ is an energy eigenstate in $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$.

⁶ Since the Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ is a complete inner-product space, the inner-product $c_n = \int_0^a [\sqrt{\frac{2}{a}} \sin(\frac{n\pi x}{a})]^* f(x) dx$, as expressed in Eqn. (4), is defined in $\mathcal{H}$. The existence of such an inner-product requires $f(x)$ to be a square-integrable function (i.e. $\int_0^a f^*(x)f(x)dx < \infty$) in $\mathcal{H}$.

⁷ The energy eigenvalue corresponding to the eigenfunction $\psi_n(x)$ is $E_n = \frac{n^2\pi^2}{2ma^2}$, where m is the mass of the particle.

⁸ Here, c_j follows Eqn. (4) for $j = n$.

⁹ Note that such an expansion of the cosine function in terms of only sine functions is not possible if the boundary points of the box were set at $x = \pm a/2$, as because for such a case, both the sine functions and the cosine functions together form a complete sets of the basis functions in $(-a/2, a/2)$.

$(\psi'_n(x) = \int_{-\infty}^{\infty} \tilde{\psi}'_n(k) \frac{e^{ikx}}{\sqrt{2\pi}} dk)$ of the state $\psi'_n(x)$, as

$$\tilde{\psi}'_n(k) = \frac{\sqrt{a/\pi}[1 - \cos(n\pi)e^{-ika}]}{n^2\pi^2 - k^2a^2} \quad (13)$$

for all $k \in \mathbb{R}$. Such a physical outcome allows one to perform the quantum (theoretical) measurement of momentum of the particle in the 1-D box and determine the corresponding position-momentum uncertainty relation $\Delta x \Delta p_x \geq \frac{\hbar}{2}$ in $\mathcal{H}_\infty$. The probability density $|\tilde{\psi}'_n(k)|^2$ corresponding to the normalized probability amplitude $\tilde{\psi}'_n(k)$ (Eqn. (13)) of the quantum measurement of momentum of the particle in the 1-D box has been plotted in figure 1 for the quantum numbers $n = 1, 2$, and 3. The secondary maxima, of course, are primarily coming from the edge-effect of the restricted geometry and can be captured within particle scattering by a standing-wave Schrödinger field [36–39]. It is clear from the figure that primary maxima of the probability density $|\tilde{\psi}'_n(k)|^2$ appear at the discrete values of the wave-number $k = k_\pm = \pm n\pi/a$ corresponding the energy eigenvalue $E_n = \frac{\hbar^2 k_\pm^2}{2m}$ except for $n = 1$ ¹¹. All the corresponding momentum eigenstates $\{\frac{e^{2\pi i j x/a}}{\sqrt{a}}; j \in \mathbb{Z}\}$, don't, of course, form an orthonormal and complete basis in $\mathcal{H}$. One can alternatively introduce a non-Hermitian momentum operator to make this set orthonormal and complete [24]. On the other hand, a bit different set of the 'free particle' states $\{\frac{e^{2\pi i j x/a}}{\sqrt{a}}\}$ with the discrete values of wave numbers $k = 0, \pm 2\pi/a, \pm 4\pi/a, \dots$, as used in Eqn. (7), is often considered in condensed matter physics as they form an orthonormal and complete set in $\mathcal{H}$. All these orthonormal and complete set of the momentum eigenstates truly don't correspond to the particle in the box of length a , but rather pick up the length $a/2$ with the energy eigenvalues $\{\frac{4j^2\pi^2}{2ma^2}\}$.

It should also be mentioned that states $\{\psi'_n(x) \in \mathcal{H}_\infty$ (Eqn. (12)) in which we want to measure momentum of the particle are neither energy eigenstates nor complete (though orthonormal by construction) in $\mathcal{H}_\infty$ for all $x \in \mathbb{R}$, and, this, of course, is not a requirement for the measurement of momentum. The momentum (observable) eigenstates in $\mathcal{H}_\infty$ must, however, be orthonormal and complete. In fact, the eigenstates $\{\frac{e^{ikx}}{\sqrt{2\pi}}\}$ of the momentum operator $\hat{p}_x = -i\hbar \frac{d}{dx}$, which appeared in the orthonormal state $\psi'_n(x)$ in Eqn. (13), are orthonormal and complete¹² in $\mathcal{H}_\infty$. Thus, quantum measurement of momentum of the particle in the 1-D box

is possible in a state $\psi'_n(x)$ with the normalized probability amplitude $\tilde{\psi}'_n(k)$ (Eqn. (13)) for the outcome $p_x = \hbar k$ and $\mathcal{H}_\infty = \mathcal{L}^2((-\infty, \infty), w(x) = 1, dx)$ becomes the appropriate Hilbert space for the same. A real measurement of the same needs the Hilbert space $\mathcal{H}_\infty = \mathcal{L}^2((-\infty, \infty), w(x) = 1, dx)$ for its theoretical explanation.

III. AN ALTERNATIVE DESCRIPTION OF THE COMPLETENESS OF THE BASIS FUNCTIONS FOR A PARTICLE IN 1-D

From the mathematical point of view, the completeness of an orthonormal set of basis vectors/functions for a general 1-D motion of a quantum particle is introduced by its properties in a number of ways, e.g. the one mentioned in Eqn. (1) as a basis expansion property and a few others, including the closure ($\sum_n |\psi_n\rangle \langle \psi_n| = \mathbb{1}$ [20]) or closure relation ($\sum_n \psi_n(x) \psi_n^*(x') = \delta(x - x')$ [4]), as mentioned in Section 1. All these forms are related to each other. The completeness of the basis functions $\{\psi_n(x)\} \in \mathcal{H} = \mathcal{L}^2((a, b), w(x) = 1, dx)$, which are the eigenfunctions of a Hermitian operator (such as the Hamiltonian), can, however, be alternatively described by a simple algebraic relation

$$\psi_m(x) = \sum_{n=0}^{\infty} \psi_n(x) \delta_{n,m} \quad (14)$$

for all $\psi_m(x)$ (say $m = 0, 1, 2, \dots$) in the orthonormal and complete set $\{\psi_m(x)\} = \{\psi_n(x)\} \in \mathcal{H}$.

Let us now see how Eqn. (14) can reach the closure relation for completeness. By virtue of the orthonormality of the basis functions in terms of the Kronecker delta and the properties of Dirac delta, we recast Eqn. (14), as

$$\int_a^b \left(\left[\sum_{n=0}^{\infty} \psi_n^*(x') \psi_n(x) \right] - \delta(x - x') \right) \psi_m(x') dx' = 0. \quad (15)$$

Since Eqn. (15) is true for any arbitrary $\psi_m(x')$ within the orthonormal and complete set of states $\{\psi_m(x')\} \in \mathcal{H}$, we have the closure relation (Eqn. (5)) for the completeness of the basis functions $\{\psi_m(x)\} = \{\psi_n(x)\} \in \mathcal{H}$. The closure relation can also be similarly derived from the basis expansion property (Eqn. (1)) of the completeness and vice versa.

We, however, are not claiming the derivation of Eqn. (5) as proof of the completeness of the basis functions, as because it is derived from Eqn. (14) where the completeness is already inbuilt. However, Eqn. (14) though looks obvious with the simple algebraic form, it is powerful enough to describe all the properties of the completeness of the basis vectors/functions.

¹¹ For the case of $n = 1$, there is only one maximum in $\psi_1(x)$, resulting in only one primary maximum of $|\tilde{\psi}'_1(k)|^2$ at $k = 0$.

¹² Here, by completeness, we mean $\int_{-\infty}^{\infty} \frac{e^{-ikx}}{\sqrt{2\pi}} \frac{e^{ikx'}}{\sqrt{2\pi}} dk = \delta(x - x')$ for all $x, x' \in \mathbb{R}$.

IV. CONCLUSIONS

To conclude, we have revisited the completeness in quantum measurement for a (quantum) particle in 1-D. We have explicitly shown an alternative derivation of the closure relation for the completeness of the energy eigenfunctions $\{\psi_n(x)\} \in \mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$ of a particle in a 1-D box of length a by the use of the Poisson summation formula. This derivation essentially, is the proof of the completeness of the energy eigenfunctions in $\mathcal{H}$. We also have shown, under the consideration of the orthonormality and completeness of the momentum eigenstates, how it is possible to physically measure momentum of the particle inside the box even if the momentum operator is not self-adjoint and not bound in the Hilbert space $\mathcal{H}$. We have also described the completeness of the eigenfunctions of a Hermitian operator in a very simple algebraic manner for a generic 1-D quantum mechanical problem. Eqns. (9), (13), & (14), and figure 1 (including the inset) are the key and novel results obtained by us.

The convergence [40] of the sine series to $f(x)$, as described in Eqn. (11), is of point-wise type for any continuous function $f(x) \in \mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$. If $f(x) \in \mathcal{H}$ becomes a piecewise continuous function with a finite amount of discontinuity at an interior point $c \in (0, a)$, then the sine series at $x = c$ would converge to the mean of the left and right-hand limits of $f(x)$ once $x \rightarrow c$ [31]. If $f(x) \in \mathcal{H}$ is a continuous function (e.g. $f(x) = (x - a)x$) and obeys the same Dirichlet boundary conditions ($f(0) = f(a) = 0$) as obeyed by $\{\psi_n(x)\}$, then the convergence of the sine series to $f(x)$, for all $x \in (0, a)$, can be shown to be of uniformly convergent type which, of course, is a special case of the point-wise type. If $f(x)$ doesn't follow only the same boundary conditions, say follows the Neumann boundary conditions (e.g. $f(x) = \sqrt{\frac{2}{a}} \cos(\frac{\pi x}{a})$) as plotted in inset-a of figure 1, then the convergence for all $x \in (0, a)$ would not be of uniform type, but rather would be of point-wise type.

The straightforward proof of the completeness (Eqn. (11)) along with the rigorous derivation of the closure relation (Eqn. (10)), as described in Section 2A for the particle in the 1-D box, doesn't directly require Dirichlet's theorem for the Fourier series. This makes our proof an alternative to the original one [2, 3]. The Fourier transform of a Dirac-comb is another Dirac-comb [30]. However, out of all the Dirac deltas (obtained after the Fourier transform), only one Dirac delta survives for a particle in a 1-D box. We have shown this (in a different way) from the Fourier series expansion and the Poisson summation formula, instead of the Fourier transform of the Dirac-comb. The only Dirac delta which survives after the Fourier series expansion leads to the derivation of the completeness closure relations for all the energy eigenfunctions (as well as proof of the completeness) for a particle in a 1-D box.

The quantum measurement of momentum of the particle in the 1-D box is not possible in the Hilbert space $\mathcal{H} = \mathcal{L}^2((0, a), w(x) = 1, dx)$, but possible in $\mathcal{H}_\infty = \mathcal{L}^2((-\infty, \infty), w(x) = 1, dx)$ where the momentum eigenstates are orthonormal and complete.

The completeness of eigenfunctions of a Hermitian operator can alternatively be described as $\psi_m(x) = \sum_n \psi_n(x) \delta_{n,m}$ for all $\psi_m(x) \in \mathcal{L}^2((a, b), w(x) = 1, dx)$.

Our alternative method of the derivation of the closure relation (Eqn. (10)), the critical answer (Eqn. (12) and figure 1) about the possibility of the quantum measurement of momentum of the particle in the box, and the simple and powerful algebraic description (Eqn. 14) of the completeness of the eigenstates of a Hermitian operator for a generic 1-D problem offer a deeper understanding of the completeness in quantum measurement.

Finally, the completeness of the energy eigenfunction was directly proved for the particle in the box [2, 3]. We believe that completeness of the energy eigenfunctions can also be directly proved for other systems, such as a 1-D simple harmonic oscillator, a hydrogen atom, a particle in a 3-D infinite square well potential, *etc.* We keep direct proofs of the same as open problems.

Acknowledgement

S. Biswas acknowledges the partial financial support of the SERB (now ANRF), DST, Govt. of India under the EMEQ Scheme [No. EEQ/2023/000788]. Useful discussions with Mr. Rhitabrata Bhattacharyya (CGST-CE, Kolkata, India), Dr. Abhiram Soori (UoH, India), Dr. Joydip Mitra (SCC, India), Dr. Saugata Bhattacharyya (VC, Kolkata, India), Prof. Koushik Ray (IACS, India), Prof. Palash B. Pal (IACS, India), Prof. Jayanta K. Bhattacharjee (IIT-Kgp, India), and Prof. Hans G. Feichtinger (UV, Austria) are gratefully acknowledged.

Author Contributions

SB framed the problem, solved the problem, and wrote the article. KBC solved the problem and wrote the article.

Data Availability Statement

No new data were created or analysed in this study.

Declarations

Conflict of Interest: The authors declare no conflicts of interest.

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